



Profile Summary

- M.Tech CSE (AI) student at IIIT Delhi with hands-on experience in GenAI systems, RAG pipelines, multi-agent workflows, backend APIs, and scalable AI software development.
- Experienced in building agentic AI systems using MCP-style tool servers, A2A agent communication, orchestrator workflows, evaluation metrics, observability, and production-ready ML pipelines.

Education

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| <b>Indraprastha Institute of Information Technology, Delhi (IIITD)</b><br>Masters of Technology CSE– CGPA : 7.71*   Current     | (2024-Expected Jun 2026)<br>Delhi |
| <b>Dr.A.P.J. Abdul Kalam Technical University Lucknow (AKTU) (UPTU)</b><br>Bachelors of Technology CSE– CGPA : 8.04   Completed | (2016-2020)<br>Ghaziabad, UP      |

Technical Skills

- Programming & CS Fundamentals:** Python, C++, Data Structures & Algorithms, Operating Systems (OS), Database Management Systems (DBMS), Computer Networks (CN), Debugging, Reinforcement Learning, Optimization, Simulation
- Backend & Systems:** FastAPI, REST APIs, Linux, System Debugging, Performance Optimization (basic),MCP-style Tool Servers, A2A Agent Communication, Agent Orchestration, Tool-calling, Trace-level Observability
- Frameworks & Tools:** Docker, Git, Software Testing, Test Automation, OR-Tools (Learning), Stable-Baselines3 (Learning), MLflow (Learning), NetworkX, SQLite, Docker, Git, FastAPI, HTTPX
- Databases:** MongoDB, FAISS, SQLite, Vector Databases
- Machine Learning/ GenAI:** PyTorch, Transformers, Model Optimization, Reinforcement Learning (Learning), Experiment Design, Evaluation Frameworks, RAG, LLM Evaluation, Prompt Engineering, Context Management, Function Call Graphs, Agentic Workflows
- Full Stack AI/MLOps:** Streamlit (Frontend), React (Learning), End-to-End ML Pipeline Development

Relevant Coursework

|                             |                         |                     |                     |
|-----------------------------|-------------------------|---------------------|---------------------|
| Graduate Algorithms         | Deep Learning           | Operating System    | Machine Learning    |
| Object Oriented Programming | Artificial Intelligence | Computer Networking | Database Management |

Experience

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| <b>Research – Microbiomes Informatics Lab, IIIT Delhi</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | (Jan 2025 – Present) |
| <ul style="list-style-type: none"><li>Engineered a Transformer-based protein language model for Genotoxin classification, achieving 98% accuracy and 0.9993 F1-score.</li><li>Built scalable data pipelines for protein sequence ingestion, preprocessing, training, and evaluation across large-scale datasets.</li><li>Applied contrastive learning, MLM pretraining, and LoRA fine-tuning to improve model performance on ~200B protein sequences.</li><li>Performed debugging, error analysis, metric-based validation, and feedback-driven improvements to enhance reliability and reproducibility.</li></ul> |                      |

Projects

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| <b>Agentic Code Intelligence Platform using MCP and A2A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | FastAPI, MCP, A2A Agents, AST, NetworkX, SQLite, Docker | (Apr 2026 – May 2026) |
| <ul style="list-style-type: none"><li>Built an MCP + A2A based agentic code intelligence for large repository understanding, generating file-level, function-level, dependency-level summaries, API-flow tracing, and function call graph analysis using orchestrator-driven multi-agent workflows with pluggable LLM support.</li><li>Designed secure MCP-style tooling for AST parsing, dependency extraction, function analysis, Git diff tracking, SQLite caching, observability, workspace sandboxing, and secret-pattern redaction for scalable and production-oriented repository intelligence pipelines.</li></ul>                                                                                        |                                                         |                       |
| <b>CineMatch: Hybrid Movie Recommendation System using SVD and LLM</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Python, TMDB API, Flask/FastAPI                         | (Mar 2026 – Apr 2026) |
| <ul style="list-style-type: none"><li>Built a two-stage recommendation pipeline using SVD-based candidate generation and LLM-based reranking on large-scale MovieLens data, improving personalization through contextual movie metadata and prompt-driven ranking.</li><li>Developed an end-to-end recommendation system integrating data preprocessing, matrix factorization, prompt engineering, backend APIs, and a frontend-backed real-time personalized recommendation workflow.</li><li>Evaluated recommendation quality using HR@K and NDCG@K, performed debugging and error analysis across retrieval/reranking stages, and improved response relevance through iterative pipeline refinement.</li></ul> |                                                         |                       |
| <b>YouTube RAG Chatbot</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | LangChain, OpenAI APIs, FAISS, Python                   | (Jan 2026 – Feb 2026) |
| <ul style="list-style-type: none"><li>Built an AI-driven retrieval and decision-support pipeline leveraging semantic retrieval, dynamic prompt orchestration, and top-K ranking strategies for context-aware response generation.</li><li>Developed a top-K retrieval system with dynamic prompt generation and OpenAI ChatModel integration to generate context-aware, coherent, and semantically relevant responses.</li></ul>                                                                                                                                                                                                                                                                                  |                                                         |                       |
| <b>Single Object Tracking using Hybrid Machine Learning Models</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PyTorch, Random Forest,ORB, SIFT                        | (Apr 2025 – May 2025) |
| <ul style="list-style-type: none"><li>Used ORB, SIFT, HOG, LBP, Linear Regression, Random Forest, OpenCV, and scikit-learn for motion compensation, feature extraction, adaptive tracking, and hybrid prediction.</li><li>Designed a robust Single Object Tracking pipeline integrating motion compensation, multiscale adaptive tracking, and feature fusion with hybrid regression models, achieving 85% IoU, 20% error reduction, and <math>R^2 = 0.92</math> predictive accuracy.</li></ul>                                                                                                                                                                                                                   |                                                         |                       |

Certifications, Achievements & Responsibility

- GATE Qualified** — Qualified GATE 2024 in Computer Science and GATE 2025 in Data Science and Artificial Intelligence.
- Udemy Agentic AI & Generative AI Certification** — Completed *AI A-Z [2026]: Agentic AI, Gen AI, Prompt Engineering and RL* from Udemy, May 2026.
- Reinforcement Learning Hackathon** — Participated in AMD AI Reinforcement Learning Hackathon by AMD, Yardi School of AI, IIT Delhi, and Unsloth.
- Kaggle Competitions** — Participated in 3 Kaggle challenge competitions involving AI/ML problem-solving and model development.

Interests

- Decision Intelligence, Optimization Systems, Reinforcement Learning, Generative AI, Scalable AI Systems, MLOps.